RUIJIA ZHANG

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EDUCATION

Johns Hopkins University **Baltimore, USA**
Ph.D. Student in Applied Mathematics and Statistics Aug. 2024 - Present
Advisor: Prof. Soufiane Hayou

The Chinese University of Hong Kong **Shenzhen, China**
B.Sc. in Mathematics and Applied Mathematics, First Class Honors Sept. 2020 - May. 2024
Advisors: Prof. Mingyi Hong, Prof. Yin Zhang

Technical University of Munich **Munich, Germany**
Exchange Student in Informatics Feb. 2023 - Oct. 2023

RESEARCH FOCUS

Reinforcement Learning for LLM, **Training Dynamics and Scaling**, **Multi-agent Systems**, **Optimization**

SKILLS

Python, MATLAB, Java, Linux, Git, PyTorch, JAX, Gurobi, COPT, vLLM

INDUSTRY EXPERIENCE

Mitsubishi Electric Research Laboratories (MERL) Cambridge, MA
Research Intern May 2026 – Aug. 2026
Project: LLM Fine-tuning for Industrial Anomaly Detection

- Fine-tuned an LLM for fault detection and severity estimation, designing a numerical sensor-to-LLM interface for multimodal industrial time-series data.
- Developed a domain-adaptation framework for transferring fault-detection models across motors, using conditional Wasserstein distributionally robust optimization to improve robustness under real deployment shifts.

PUBLICATIONS & PREPRINTS

Ruijia Zhang and Soufiane Hayou. **Width and Depth Limits Commute During Training in Linear Residual Networks.** *In submission to ICLR 2027.*

Ruijia Zhang, Dehong Liu, and Yebin Wang. **Sparse Target Calibration for Domain Adaptation: A Conditional Distributionally Robust View.** *In submission to ICLR 2027.*

Ruijia Zhang, Jiacheng Zhu, Hanqing Zhu, and Laixi Shi. **Geometry-Preserving Orthonormal Initialization for Low-Rank Adaptation in RLVR.** *ICML 2026*

Ruijia Zhang, Zhenglin Qi, Yue Wu, Xiangyu Zhang, Yanxun Xu. **POLAR: A Pessimistic Model-based Policy Learning Algorithm for Dynamic Treatment Regimes**. *Journal of the American Statistical Association (JASA)*

Ruijia Zhang, Siliang Zeng, Chenliang Li, Alfredo Garcia, and Mingyi Hong. **Understanding Inverse Reinforcement Learning under Overparameterization: Non-Asymptotic Analysis and Global Optimality.** *AISTATS 2025*

Ruijia Zhang, Mingxi Lei, Meng Ding, Zihang Xiang, Jinhui Xu, Di Wang. **Improved Rates of Differentially Private Nonconvex Strongly Concave Minimax Optimization via Gradient Differences.** *AAAI 2025*

TECHNICAL REPORTS

On the Scaling of PEFT: Towards Million Personal Models of Trillion Parameters

MinT: Managed Infrastructure for Training and Serving Millions of LLMs.

RESEARCH PROJECTS

Joint Width–Depth Scaling Laws for Neural Network Training
Graduate Researcher at Johns Hopkins University Feb. 2026 – Sept. 2026
Advisor: Prof. Soufiane Hayou

- Proved that, under $1/\sqrt{\text{depth}}$ residual-branch scaling and clipped gradient descent, trained outputs converge in L^2 to the same limit along arbitrary joint width–depth growth paths, for any fixed number of training steps.
- Derived from training dynamics the width (N)–depth (L) scaling law $\boldsymbol{N \propto L^2}$, which is empirically observed across modern LLM families.

Geometry-Preserving Orthonormal Initialization for Low-Rank Adaptation in RLVR
Joint work with researchers at Meta Superintelligence Labs and xAI Sep. 2025 – Feb. 2026
Collaborators: Dr. Jiacheng Zhu, Dr. Hanqing Zhu, and Prof. Laixi Shi

- Identified why SVD-based LoRA initializations such as PiSSA and MiLoRA become unstable under RLVR, showing that their failures arise from amplified gradients and excessive policy drift.
- Proved that orthonormal initialization minimizes the gap to full fine-tuning. Proposed geometry-preserving LoRA-RLPO and LoRA-RLMO.
- Stabilized RLVR training and improved reasoning performance by **18.3%** over standard LoRA on a 7B model.

POLAR: Pessimistic Multi-Stage Offline Reinforcement Learning
Graduate Researcher at Johns Hopkins University May 2024 – Oct. 2024
Advisors: Prof. Zhenglin Qi and Prof. Yanxun Xu

- Developed POLAR, a model-based algorithm for history-dependent, multi-stage offline RL through iterative function approximation.
- Incorporated pessimism through transition-uncertainty penalties, enabling robust policy learning under partial coverage while relaxing standard positivity assumptions.
- Established finite-sample suboptimality and computational guarantees, achieving a **12% higher policy value** than the strongest baseline under limited data coverage.

Neural Inverse Reinforcement Learning for Robotic Control
Undergraduate Research Intern at University of Minnesota May 2023 – Sep. 2023
Advisors: Prof. Mingyi Hong and Prof. Alfredo Garcia

- Developed the first single-loop maximum-likelihood IRL algorithm for learning neural reward functions and control policies from expert demonstrations, eliminating nested policy optimization.
- Evaluated in a single-demonstration setting on Hopper, HalfCheetah, Walker, and Ant, achieving the best non-expert return on three of four MuJoCo tasks and **99.5% of expert return** on HalfCheetah.
- Established an $\mathcal{O}(\epsilon^{-8})$ finite-time complexity bound and global recovery of the optimal reward function and policy under overparameterized ReLU models.

Differentially Private Minimax Optimization via Gradient Differences
Visiting Student at King Abdullah University of Science and Technology Dec. 2022 – Apr. 2023
Advisor: Prof. Di Wang

- Provided the first utility analysis of differentially private gradient descent–ascent for nonconvex–strongly-concave minimax optimization without the Polyak–Łojasiewicz condition.
- Proposed PrivateDiff to reduce privacy-induced gradient variance, improving the squared-gradient utility rate from $\widetilde{O}\Big(\sqrt{d}/(n\varepsilon_{\text{DP}})\Big)$ to $\widetilde{O}\big(d^{2/3}/(n\varepsilon_{\text{DP}})^{4/3}\big)$.
- Improved AUC by **7.7 points** over DP-SGDA on MNIST under $\varepsilon = 0.5$ privacy and consistently reduced temporal-difference loss across reinforcement-learning benchmarks.

AWARDS

ASA Nonparametric Statistics Section Best Student Paper Award 2026
International Conference on Continuous Optimization Student Grant 2025

TEACHING EXPERIENCE

Time Series Analysis (Spring/Fall 2026). Data Science Methods for Large-Scale Graphs (Spring 2025).